

Most image sharpening software tools work by applying something called an 'unsharp mask,' which actually acts to sharpen an image. The tool works by exaggerating the brightness difference along the edges within an image. Note that the sharpening process is not able to reconstruct the ideal image, but it creates the appearance of a more pronounced edge.

The command used for sharpening an image in MATLAB is:

```
B = imsharpen(A)
```

It returns an enhanced version of the grayscale or the true-colour (RGB) input image A, where image features such as edges have been sharpened using the unsharp mask-ing method.

B = imsharpen(A, Name, Value,...) sharpens the image using name-value pairs to control aspects



Fig. 12: Modified image

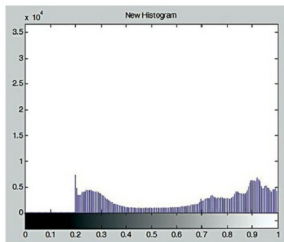


Fig. 13: Histogram of modified image shown in Fig. 12



Fig. 14: Original image



Fig. 15: Image after thresholding

of unsharp masking.

Let us see the use of imsharpen function:

```
>> a=imread('Image_sharpen.jpg');
>>imshow(a)
```

Fig. 16 shows the image to be sharpened (Image_sharpen.jpg).

Following commands sharpen the image using the simple imsharpen command:

```
>> b=imsharpen(a);
>>figure, imshow(b)
```

Fig. 17 shows the resultant sharpened image.

You can specify 'radius' and 'amount' parameters in the imsharpen function as given in the example below:

```
>> b=imsharpen(a, 'Radius',
4, 'Amount', 2);
>>figure, imshow(b)
```

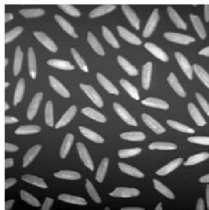


Fig. 16: Image to be sharpened (Image_sharpen.jpg)

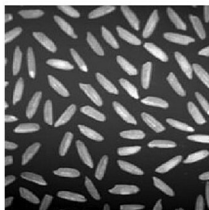


Fig. 17: The resultant sharpened image

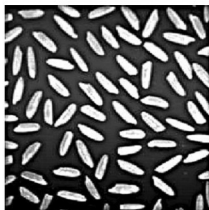


Fig. 18: Resultant image after specifying 'radius' and 'amount' parameters in the imsharpen function

Fig. 18 shows the resultant image.

As you can see, image in Fig. 18 is much sharper than the image in Fig. 17, which, in turn, is much sharper than the image in Fig. 16.

To be continued next month